

[12] Developing a Computer-Based Tutor Utilizing Generative Artificial Intelligence (GAI) and Retrieval-Augmented Generation (RAG)

- **Authors:** Youngjin Lee
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1-Minute Executive Takeaway

Key Finding: Direct EAIT precedent confirming that RAG is essential for curriculum grounding and eliminating AI inaccuracies in higher education.

Core Research Objective

To design, implement, and evaluate an intelligent computer-based tutor utilizing generative AI and RAG to support personalized statistics education while curbing hallucinations.

Methodology & Architecture

Integrated external textbook and curriculum resources into a RAG framework, evaluating student interaction logs, conceptual learning gains, and factual response accuracy.

Key Empirical Findings & Evidence

Demonstrated that grounding LLM responses in curricular texts significantly enhances conceptual accuracy, mitigates hallucination, and fosters student trust in automated tutoring systems.

Direct Relevance to Our Kwiz Paper

Crucial precedent published in EAIT (our target journal). Lee (2025) proves EAIT's active interest in educational RAG. Our work directly complements Lee by addressing the instructor-facing assessment authoring counterpart (generating verified questions for the LMS Question Bank).